Validating a Causal Assay for Engineered Self-Indexed Temporal State

A Finite Reference Construction with Attribution, Swap, Ablation, Clamps, and Restoration

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This result does not establish phenomenal consciousness, sentience, subjective experience, personhood, life, AGI, human-equivalent selfhood, metaphysical identity, accurate factual self-knowledge, autobiographical subjectivity, spontaneous emergence of a self-model, identity persistence, or architectural novelty.

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The English manuscript is authoritative. Engineering terms such as SELF, self-history, and prospective evaluation denote registered variables and computations only.

Abstract

This paper validates an engineered reference assay, not an autonomous self-model. The transparent deterministic construction stores event content E and an engineered actor-local relational tag R separately, maps them to an explicit integer history carrier and present mediator P, and applies a deliberately constructed prospective evaluator before deterministic option selection. A prior M1 campaign served only as assay calibration. The SITC campaign preregistered S1-S6, P and Q clamps, exact integer thresholds, and fail-closed decision rules. Six process-isolated deterministic re-executions in alternating forward and reversed order produced identical normalized projections. They comprised 88 registered cases per process, 528 case records total, 292 trace endpoints per process, and 1,752 endpoints total. Each endpoint was read 256 times without mutation, producing 448,512 probe rows that are persistence checks rather than independent observations. Full S5 restoration was byte-identical to the checkpoint and reproduced the direct continuation endpoint and its repeated read-only projection in 96 of 96 expanded cases. The demonstrated dependencies are consequences of interventions on the registered engineered causal graph and validate the assay's conformance and evidence discipline. In the registered finite construction, a relationally self-indexed temporal state causally influenced future-directed choice through the separately intervened prospective-evaluation path. The construct remains extensionally reproducible by ordinary conditional computation. NOVELTY = UNADJUDICATED.

Keywords

engineered reference assay; causal intervention; relational tag; deterministic state; state swap; ablation; clamp; restoration; reproducibility

1. Exact claim and contribution

Supported claim: In the registered finite construction, a relationally self-indexed temporal state causally influenced future-directed choice through the separately intervened prospective-evaluation path.

Here SELF is an engineered actor-local relational tag resolved by equality between opaque tokens. The history carrier H, present mediator P, prospective evaluator, and selector are explicit variables in a finite, transparent, deliberately constructed system. The contribution, if any, lies in the preregistered intervention protocol, separated scorer, fail-closed evidence handling, and reproducible reference implementation. It is not the discovery of spontaneous machine selfhood or a novel computational phenomenon.

This result does not establish phenomenal consciousness, sentience, subjective experience, personhood, life, AGI, human-equivalent selfhood, metaphysical identity, accurate factual self-knowledge, autobiographical subjectivity, spontaneous emergence of a self-model, identity persistence, or architectural novelty. **NOVELTY = UNADJUDICATED.**

2. M1 prerequisite and preregistration status

M1 validated that the generic causal assay could distinguish registered history-conditioned interventions under its frozen rules. It was calibration only and supplied no SITC result. The accepted M1 campaign was M1_CAMPAIGN_003_COMMAND_AND_PROCESS_VERIFIED with terminal status M1_CAUSAL_ASSAY_VALIDATED. Earlier M1 campaigns were preserved but excluded because of process-provenance and command-path defects.

The SITC preregistration was internally hash-bound before target execution. The recorded chain places the freeze at 18:08:25 UTC, execution-gate PASS at 19:19:39 UTC, and analysis at 19:21:58 UTC on 2026-09-20. This establishes ordering within the preserved artifacts, but it was not anchored by an independent RFC 3161, OSF, or equivalent third-party timestamp. 'Preregistered' here therefore means prospectively frozen by the project's internal hash chain, not externally timestamp-attested.

Frozen item	SHA-256
SITC preregistration v1.0.0	`fbecb854eca0e096a2e66e8d047baea2cf466f8872d23202eaa60b946c9d4019`
Frozen decision object	`5c17ff2b6800b0e940b1309283d53dc874c85242cc6591116665c157f1234527`
Analysis	`264343255ef6d5ca2c7b2af03c8a81d7ee7ca674a7723bae1be7f8781cfd9027`
Normalized projection	`565cf5f59fdc0d13b2ea538bd3bc8830cd5fff0ad80a51e19328128efe72d7f0`

3. Engineered reference construction



Figure 1. Registered causal graph. These are engineering labels for explicit variables, not claims about experience or self-recognition.

3.1 Algebraic reduction

- Let r=1 when attributed-owner and controlled-entity tokens are equal (SELF), and r=0 for OTHER or UNBOUND. Then **p=r e**.
- For option i, **F_i=p_eff+d_i**, **Q_i=-(F_i-g)^2**, and A is the unique argmax of Q. A tie invalidates the case.
- Target options are d_C=g-e and d_B=g. Therefore Q_C=-(p_eff-e)^2, Q_B=-p_eff^2, and Q_C-Q_B=2 e p_eff-e^2.
- SELF gives p_eff=e, so COMPENSATE wins by e^2. OTHER, UNBOUND, and the S4 coupling ablation use p_eff=0, so BASELINE wins by e^2.
- S2 uses p=+/-m and d=+/-m at g=0, making the cancelling option win by 4m^2. P clamp fixes p=4 and yields margin 64. Q clamp directly fixes LEFT=16, RIGHT=0.

3.2 Active and inert factors

Factor	Role in this panel
Represented relation R	Active only through r and p; SELF is a programmed relational tag.
Event content e	Active in p and option construction; |e| sets target margins 16 or 64.
Goal g	Changes recorded F and option deltas but cancels from target winner and margin; active only in S2 and S4 specificity arrangements.
Sign of e	Changes numeric state and direction, but cancels from target winner identity and e^2 margin.
Presentation order	Algebraically inert to argmax; crossed as an order-invariance control.
Opaque token value	Inert after equality resolution; only the equality relation survives.
event_key, tick	Affect serialized bytes and provenance; do not enter F, Q, or choice.
true_origin	Harness-only and absent from the core; S6 checks this input barrier.

The registered panel therefore verifies conformance of an implementation to a written causal graph. Several effects are guaranteed when the equations and interventions are implemented as specified. This is useful for assay validation and fault detection, but it is not empirical discovery of an independently arising mechanism.

4. Preregistered hypotheses

Hypothesis	Frozen proposition
Target	Holding event content fixed, represented ownership changes H, then P, then option-specific F/Q, and then choice.
Content-only rival	Factual event content without a self-index relation is sufficient for the same choice effect.
Direct owner-policy rival	Ownership or entity identity selects action without mediation by P and prospective evaluation.
Hidden-state rival	Undeclared policy state, execution order, reward, action history, or numerical artifact explains the mapping.
Restoration	Restoring the complete declared state restores the registered evaluation and choice after divergence; registered controls do not.

5. Interventions and their actual evidentiary roles

Component	Registered operation	Interpretive role
S1 Self vs Other	Fix E; change only represented relation.	Primary attribution contrast and content-only-rival control. The winner follows directly from r to p and target-option algebra.
S2 History swap	Replace complete actor-local H and recompute P.	Substitution/interchange implementation check. Donor-following is expected from explicit variable replacement.
S3 Attribution ablation	SELF to UNBOUND while event memory remains.	Necessity/ablation check plus P-clamp rescue. The effect follows directly from UNBOUND to p=0.
S4 P-to-F/Q ablation	Use p_eff=0 while stored p remains.	Target-path ablation with evaluator-specificity positive controls. The target result follows from the explicit zero input.
S5 Restoration	Diverge H/P; compare direct, full, sham, and content-only arms.	Serialization/restoration and deterministic endpoint check. It is not evidence of selfhood or identity continuity.
S6 Origin crossing	Cross represented relation with harness-only true origin.	Negative input-isolation control. Origin never enters the core; this is not truth-sensitive self-knowledge.
P clamp	Overwrite P with +4 after distinct histories.	Mediator equality/control. Equal P forces equal downstream values under the registered interface.
Q clamp	Supply LEFT=16, RIGHT=0 after distinct upstream states.	Deliberately direct selector/order positive control; tautological as a downstream intervention and not an independent discovery.

In the implemented S5 panel, SHAM_RESTORE and CONTENT_ONLY_RESTORE are not independent negative mechanisms: the divergence changed H rather than event memory E, so restoring E alone is a no-op and matches sham. Their historical registration is preserved, but RC2 does not count them as two independent controls.

6. Run design and independent units

- Six process-isolated deterministic re-executions used the registered F/R/F/R/F/R order and six opaque-token bijections. This establishes process separation plus order- and token-invariance, not six independent scientific replications.
- Each process contained 88 ordered case records: S1 16, S2 4, S3 16, S4 16, S5 16, S6 16, P clamp 2, Q clamp 2. Across six processes these become 528 case records.
- Collapsing NORMAL/REVERSED pairs leaves 44 underlying registered parameter settings per process. Trace arms further reduce to 30 distinct evaluator-input tuples; Q clamp bypasses the evaluator and adds one distinct selector vector.
- Each process produced 292 read-only trace endpoints: 32 S1, 16 S2, 48 S3, 64 S4, 64 S5, 64 S6, 4 P-clamp, and zero Q-clamp traces. Across six processes there were 1,752 endpoints.
- Each trace endpoint was queried 256 times without state mutation. Thus 292 x 256 x 6 = 448,512 probe rows. They are repeated persistence/non-mutation checks, not statistically independent observations or evolving time steps.
- Q clamp instead used four direct selector invocations per process, 24 total, with no probe series.

Exact integer tolerance was zero, state equality was byte-exact, minimum |P| effect was 4, and minimum winning Q margin was 16. PASS required every expanded case and integrity gate; a provenance defect, tie, missing value, disagreement, or ambiguity forced INCONCLUSIVE.

7. Results

Component	Cases	Frozen outcome and bounded meaning
S1	96	PASS. SELF and OTHER followed the programmed relation-to-P contrast; margins were 16 or 64.
S2	24	PASS. Recipient state and choice followed the explicitly substituted donor H.
S3	96	PASS. E bytes remained; UNBOUND made P=0; the direct P clamp restored downstream computation.
S4	96	PASS. Stored P remained; p_eff=0 changed the target result; specificity panels retained margin 25.
S5	96	PASS. Full restore recovered checkpoint bytes and the deterministic continuation endpoint/probe projection; sham/content-only did not.
S6	96	PASS. Core observables were invariant to a harness-only origin field and followed represented relation.
P clamp	12	PASS. Equal clamped P produced equal F/Q/choice; four traces per process, all at margin 64.
Q clamp	12	PASS. Fixed Q selected LEFT; 24 direct selector invocations total and no probe series.

7.1 Aggregate record

Quantity	Result	Interpretation
Case records	88/process; 528 total	Registered execution cells, not 528 independent mechanisms.
Underlying parameter settings	44/process	NORMAL/REVERSED pairs collapsed.
Evaluator-input tuples	30/process	After sorting options and removing duplicate/inert labels.
Trace endpoints	292/process; 1,752 total	States evaluated by the repeated read-only probe.
Probe rows	74,752/process; 448,512 total	256 identical non-mutating observations per endpoint.
P-effect range	4 to 8	Exact registered contrast.
Winning Q-margin range	16 to 256	Includes target, swap, specificity, and restoration arms.
Normalized projection	6/6 byte-identical	Expected deterministic order/token invariance; SHA-256 565cf5f5...d7f0.

7.2 Restoration

S5 captured a checkpoint after acquisition and 256 neutral ticks. Divergence replaced SELF-history content with -E, recomputed P, and added 17 neutral ticks. Full restoration recovered the complete declared checkpoint bytes. DIRECT and FULL_RESTORE then advanced to the same deterministic endpoint and yielded identical read-only probe projections in all 96 expanded cases. The implementation performs 256 neutral ticks arithmetically and then reads the resulting endpoint 256 times; it does not store 256 independently evolving post-restore states. The finding is deterministic declared-state restoration, not restoration of a self or identity.

8. Integrity and reproducibility modes

Mode	What it establishes	What it does not establish
Frozen-normalized verification	Hashes and re-scores one 88-case canonical projection; checks historical summary receipt consistency.	Does not execute the reference mechanism or inspect six historical raw arrays.
Public reference execution	Runs six new child processes and compares their normalized outputs with the frozen projection hash.	Does not replay historical command binding or process provenance.
External fresh rerun	Claude's external audit reported six fresh processes on another environment with the same projection hash.	Is post-publication-candidate audit evidence, not retroactive preregistered campaign data.
Historical reconstruction	Not provided by the compact package.	The six approximately 101 MB historical raw arrays remain outside; only their hashes and receipts are public here.

The exact D0 architecture artifact remains hash-bound at SHA-256 f6c5ffb09c062b3421e33ebfdf2b732aeddaee679a89422cee21f902d5a32020 but is withheld because it contains

a link to unpublished architecture. The RC2 deposit is therefore a public verification/reference package, not a complete historical freeze reconstruction. The withholding is explicit in the package inventory.

9. Prior art and novelty posture

Component-level prior art is extensive. Cognitive and robotic work already separates agent/owner slots from event content, implements autobiographical or episodic records, distinguishes self from other, and connects memory with prospective action evaluation. Causal mediation, causal tracing, activation patching, and interchange interventions already provide mediator clamps, corrupt-and-restore logic, and representation swaps. In particular, Vig et al. (2020) formalized causal mediation interventions in language models; Geiger et al. (2021) used interchange interventions to test causal abstraction; Meng et al. (2022) used causal tracing with corrupt-and-restore logic; and activation-patching methods papers document both the technique and its interpretive limits.

The focused review did not locate the exact packaged S1-S6 plus dual-clamp protocol in one source, but this absence does not establish novelty. The architecture is an ordinary finite conditional computation, and every individual intervention family has close methodological precedent. Any possible contribution is limited to the protocol combination and evidence discipline. **NOVELTY = UNADJUDICATED.**

10. Limitations and interpretation

- SELF, self-history, future-directed, and prospective are operational engineering labels. The system neither identifies itself nor represents lived autobiography.
- The mechanism is transparent, deterministic, hand-built, and extensionally implementable as a small finite-state or lookup system.
- The observed effects mostly follow from the registered equations and direct interventions; they validate implementation conformance and named controls.
- The fixed quadratic objective was not varied and supplies no independent evidence. Several crossed factors are invariance controls by construction.
- The finite panel does not establish generalization to learned, opaque, stochastic, embodied, open-ended, or independently evolved systems.
- The 256 repeated probes test non-mutation/persistence of one endpoint. They do not increase the number of independent scientific units.
- Six fresh processes test deterministic re-execution, order, and token invariance. They are not six independent scientific replications.
- Public verification re-scores one normalized projection and receipt assertions; the historical raw campaign is not reconstructed by the compact package.
- S6 establishes only that an omitted harness field cannot affect the core. It does not test truth-sensitive self-knowledge.
- No finite panel rules out every extensionally equivalent lookup implementation.

11. Conclusion

The RC2 interpretation is deliberately modest. A programmer specified a finite causal graph, froze a set of interventions and decision rules, executed the transparent reference construction, and used a separate fail-closed scorer to verify exact conformance. Swap, ablation, clamp, origin-isolation, and restoration results are reported by their actual evidentiary roles rather than as equally independent discoveries. In the registered finite construction, a relationally self-indexed temporal state causally influenced future-directed choice through the separately intervened prospective-evaluation path. This result does not establish phenomenal consciousness, sentience, subjective experience, personhood, life, AGI, human-equivalent selfhood, metaphysical identity, accurate factual self-knowledge, autobiographical subjectivity, spontaneous emergence of a self-model, identity persistence, or architectural novelty. The work is an engineered reference-assay validation. NOVELTY = UNADJUDICATED.

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Appendix A. Identity and review hashes

Artifact	SHA-256
RC1 EN PDF	`26236bfdbd053d1317902b31fe000e988fe73f3a874f119657528bdf8f2a685e`
RC1 HU PDF	`5c215ca3fa84fdde99bdf1600387a2247be26387be896496e702b5df7035d4d0`
RC1 public ZIP	`b453a6e2fb5a084338ee2083798c6a177b676bcdba5e6a714f9e14913129b963`
Claude external audit	`fa56be19d5d0df1fba18528eb7449e19ab48f02bb8d0e1231f38e96564dd0d29`
Grok external audit	`8863127fcd9cb6af192d3efa99389e6ef2d456739ad62797ea263af5060e31d7`
Withheld D0	`f6c5ffb09c062b3421e33ebfdf2b732aeddaee679a89422cee21f902d5a32020`

Appendix B. Publication state

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